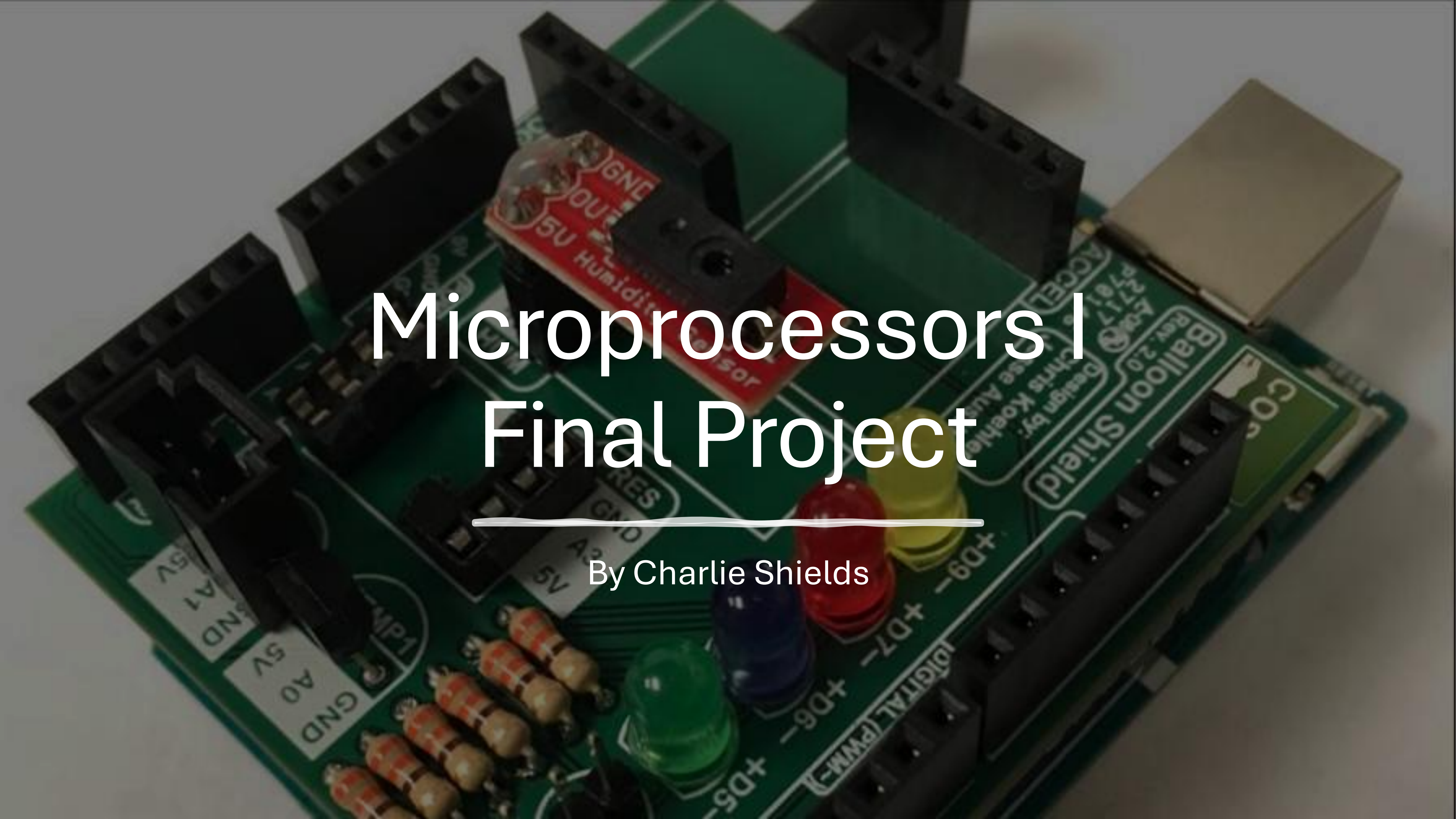


Microprocessors I Final Project

By Charlie Shields

Current Design Analysis

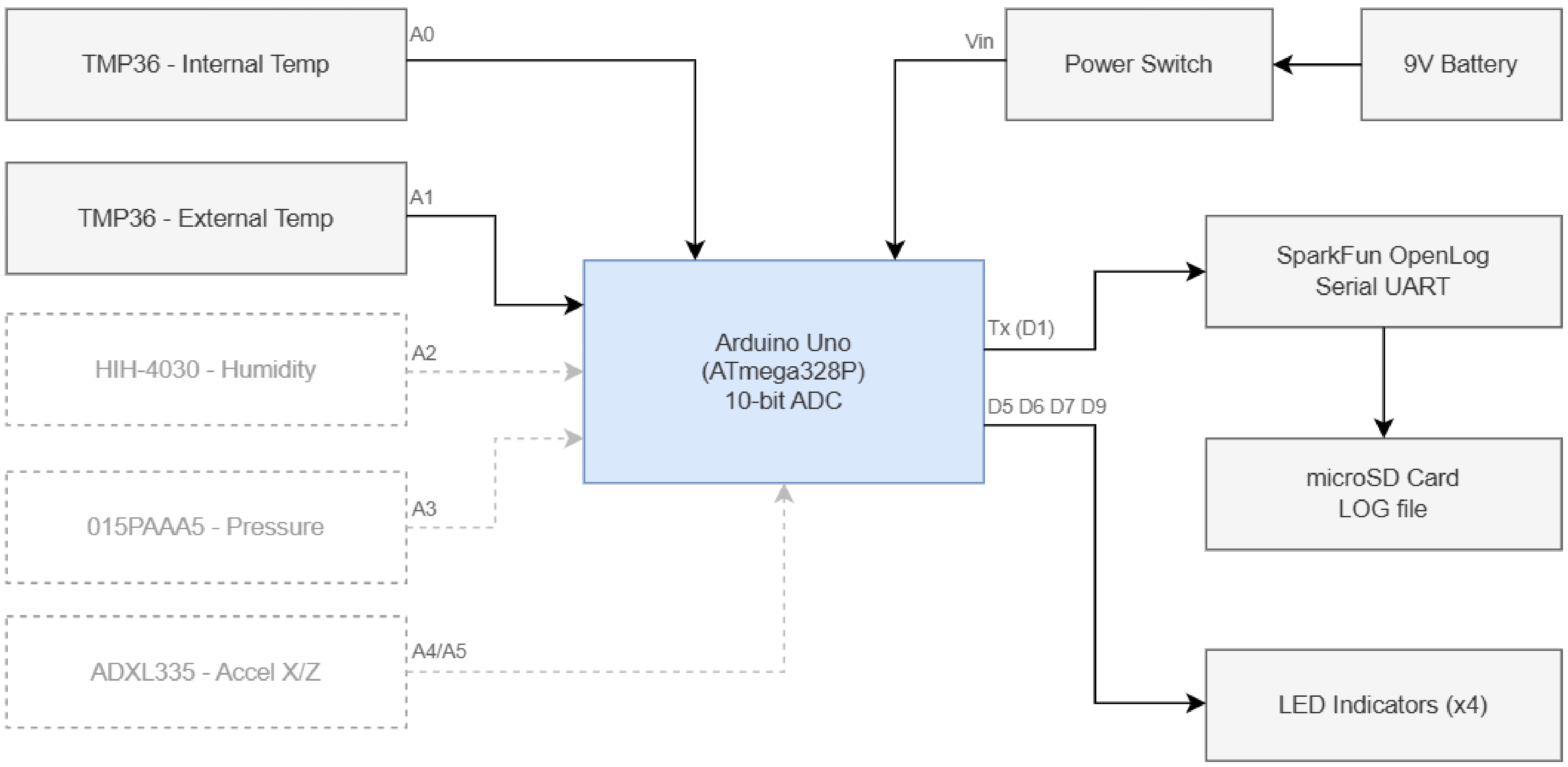
Available:

- Arduino UNO
- TMP36
- LEDs
- Data Logger
- Battery
- Power Switch
- Resistors

Obsolete:

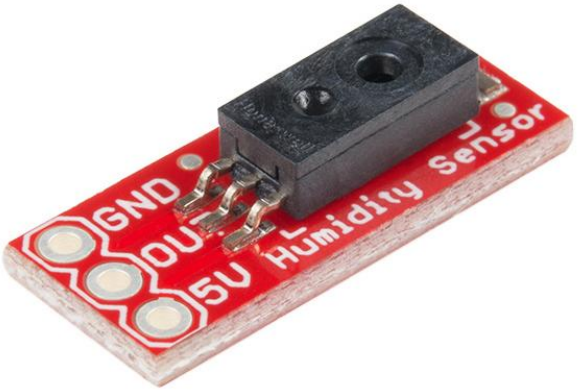
- Humidity Sensor
- Pressure Sensor
- Accelerometer

DemoSat System Block Diagram - Phase 1



Humidity & Temperature Sensors

HIH-4030



Quantity	Price
250+	$53.630

Price for: Each

1000

Add to Cart

Minimum: 1000 Multiple: 1000

$53,630.00

Manufacturer Standard Lead Time: 15 week(s)

ASDXACX015PAAA5

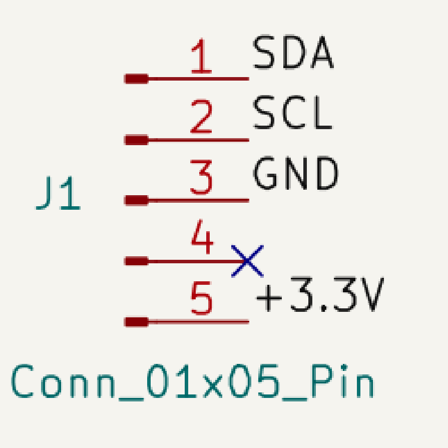
Bulk

QUANTITY	UNIT PRICE	EXT PRICE
25	$194.88120	$4,872.03

Manufacturer Standard Lead Time 17 Weeks

Adafruit MS8607 Breakout

- Pressure
- Humidity
- Temperature



Google:

Typically $15

Electromaker $14.95
Adafruit Ms8607 Pressure Humidity Temperature...
In stock online
4.4/5 · Delivery by Wed $7.99 · 30-day returns

Paradisetronic.com $22.00
Adafruit MS8607 Pressure Humidity Temperature...
In stock online

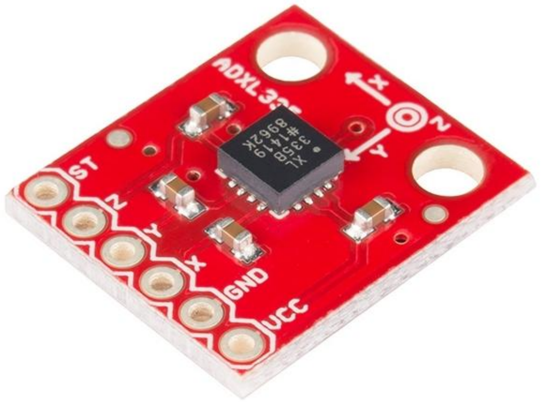
Adafruit Industries $14.95
adafruit-Adafruit MS8607 Pressure Humidity...
Out of stock online
4.9/5 · Free delivery on orders $250+ · Delivery $14.45 · 30-day returns

DigiKey:

Manufacturer Standard Lead Time 4 Weeks

Accelerometer

SparkFun ADXL335 Breakout

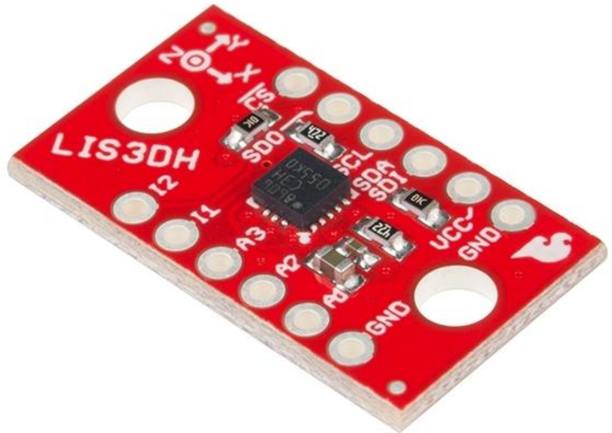


$22.27

- 1 + Backorder

BACKORDER STOCK/DISCOUNTS

SparkFun LIS3DH Breakout

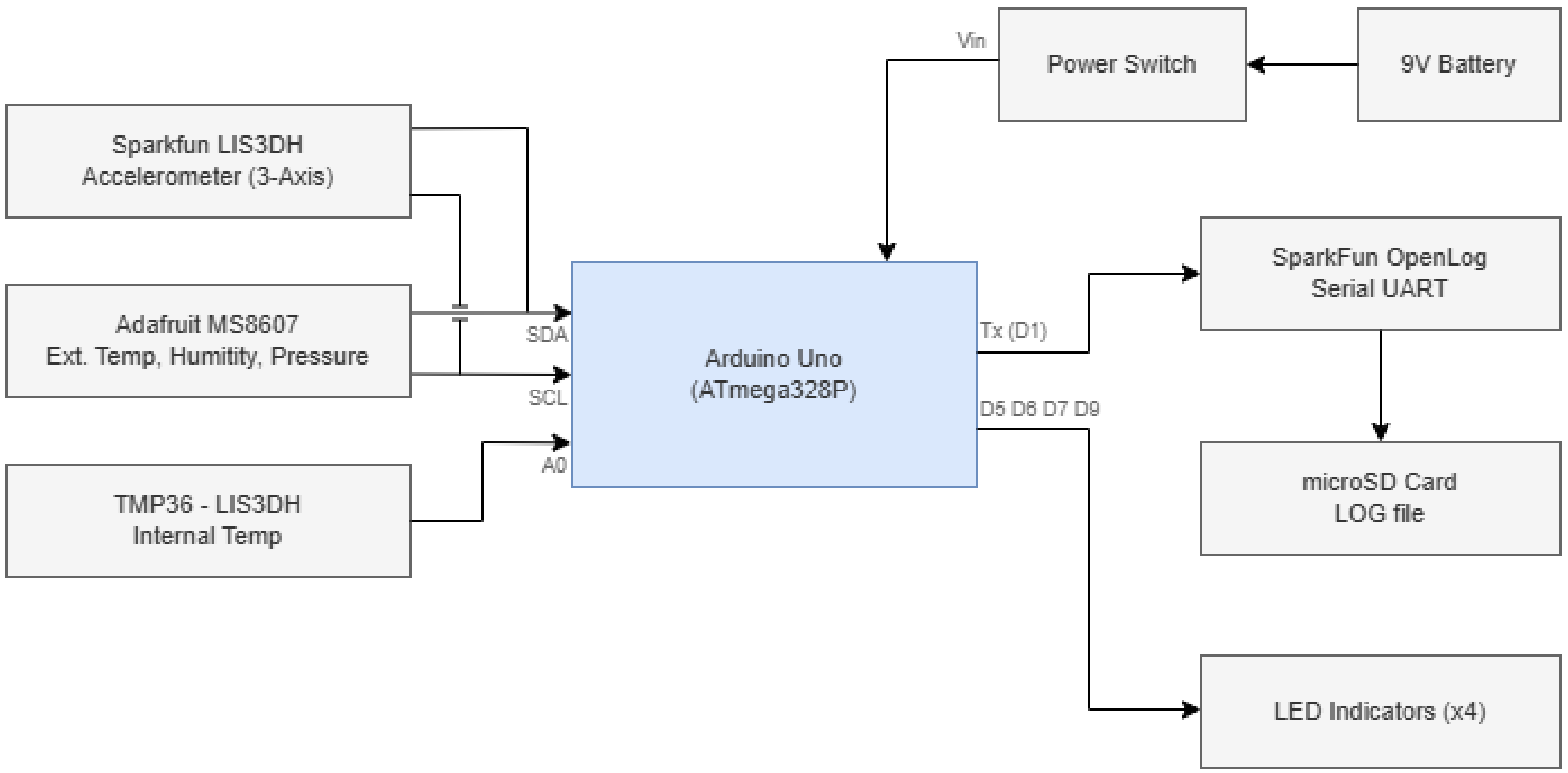


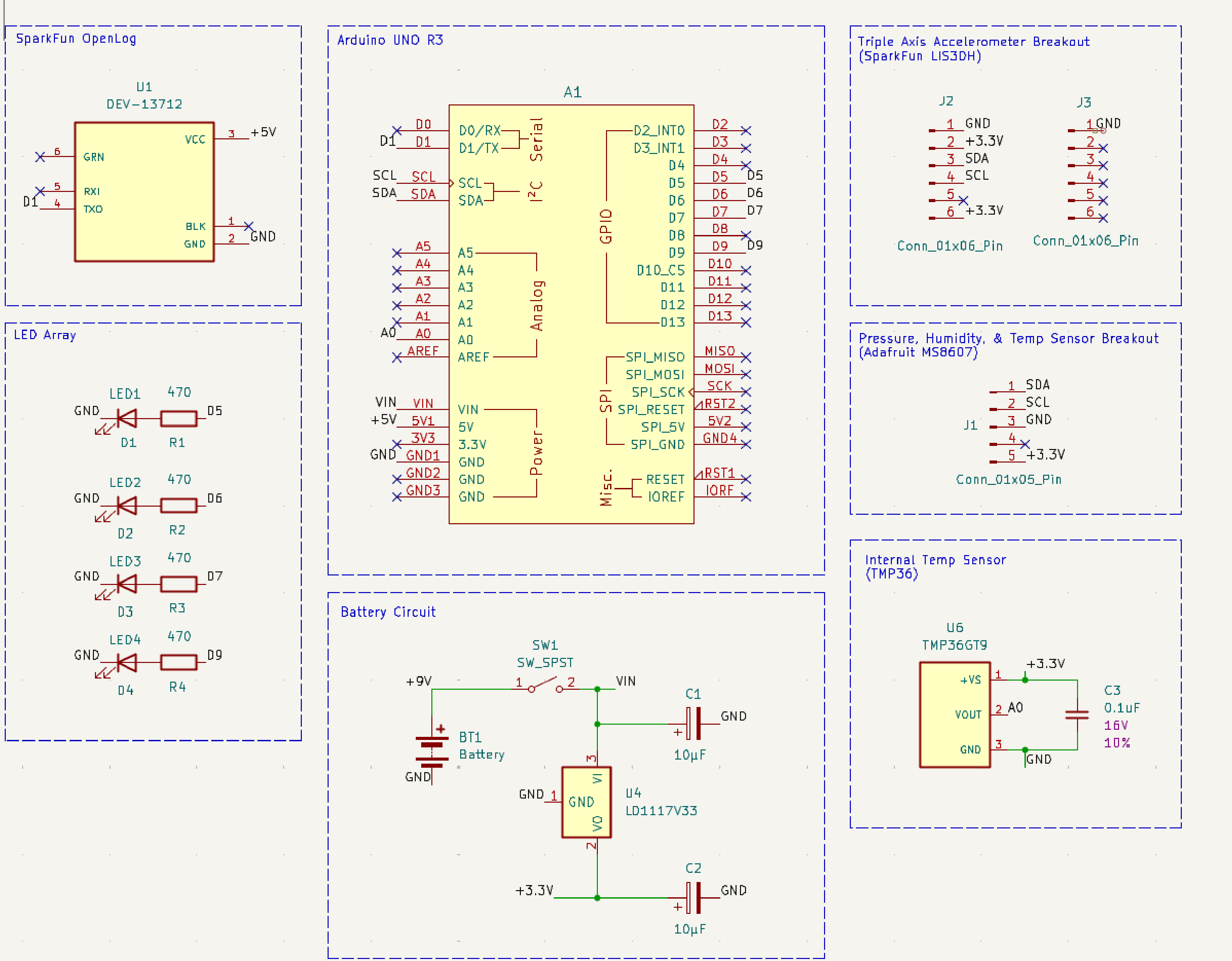
$7.75

- 1 + Add to Cart

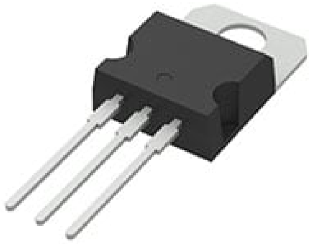
IN STOCK STOCK/DISCOUNTS

DemoSat System Block Diagram - Phase 2

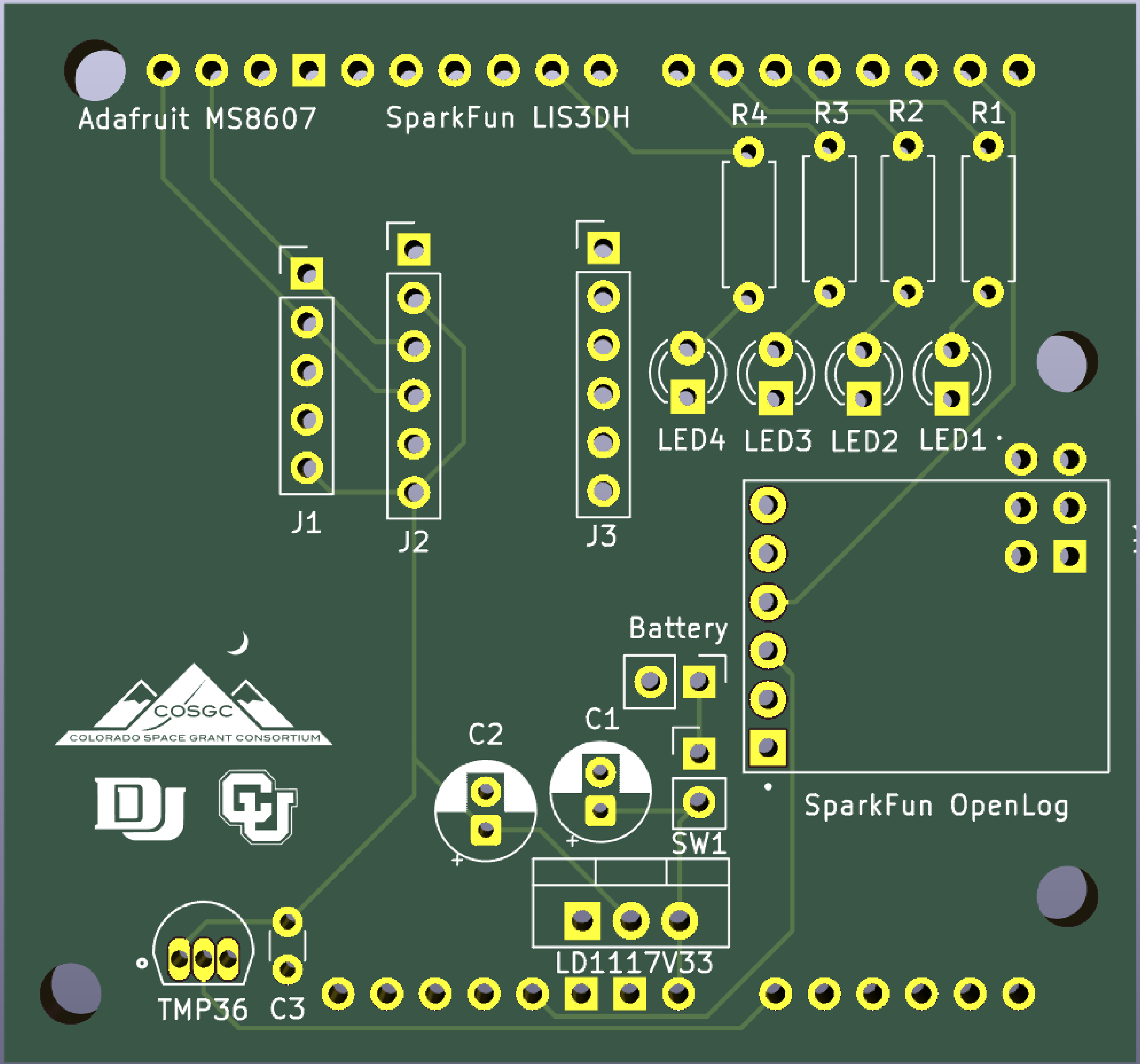


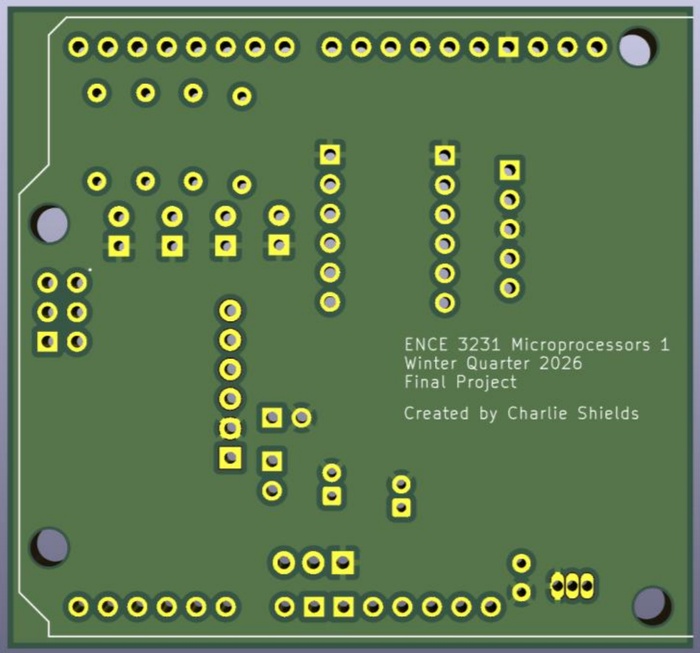


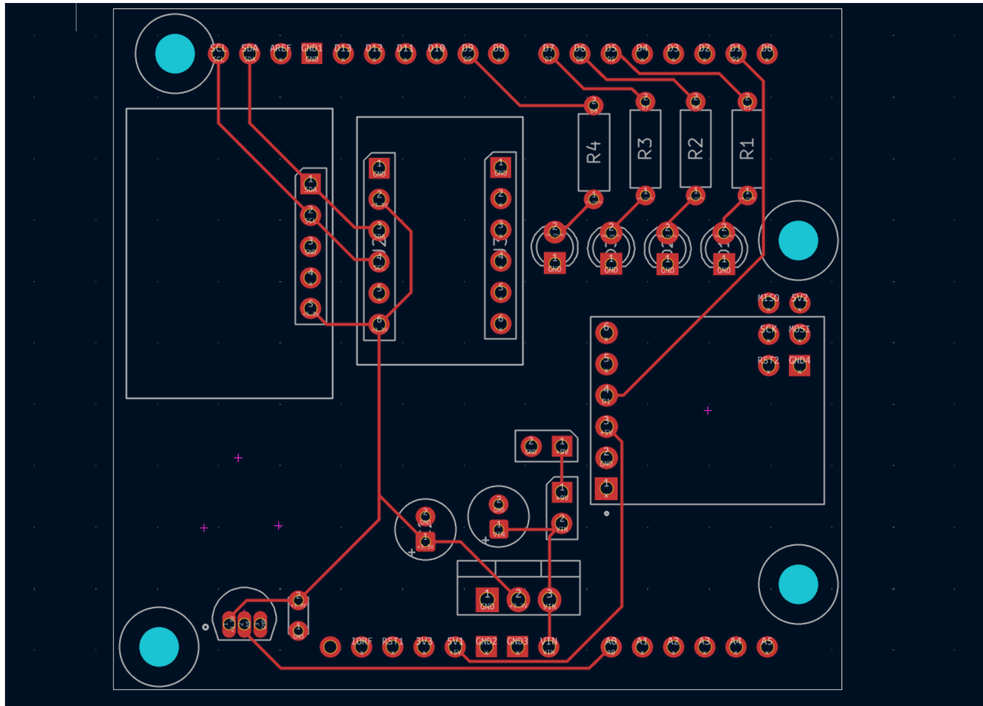
LD1117V33C



TMP36


Adafruit MS8607
SparkFun LIS3DH
R4
R3
R2
R1
LED4
LED3
LED2
LED1
J1
J2
J3
Battery
C2
C1
SW1
SparkFun OpenLog
COSGC
COLORADO SPACE GRANT CONSORTIUM
TMP36
C3
LD1117V33


ENCE 3231 Microprocessors 1
Winter Quarter 2026
Final Project
Created by Charlie Shields


R4
R3
R2
R1

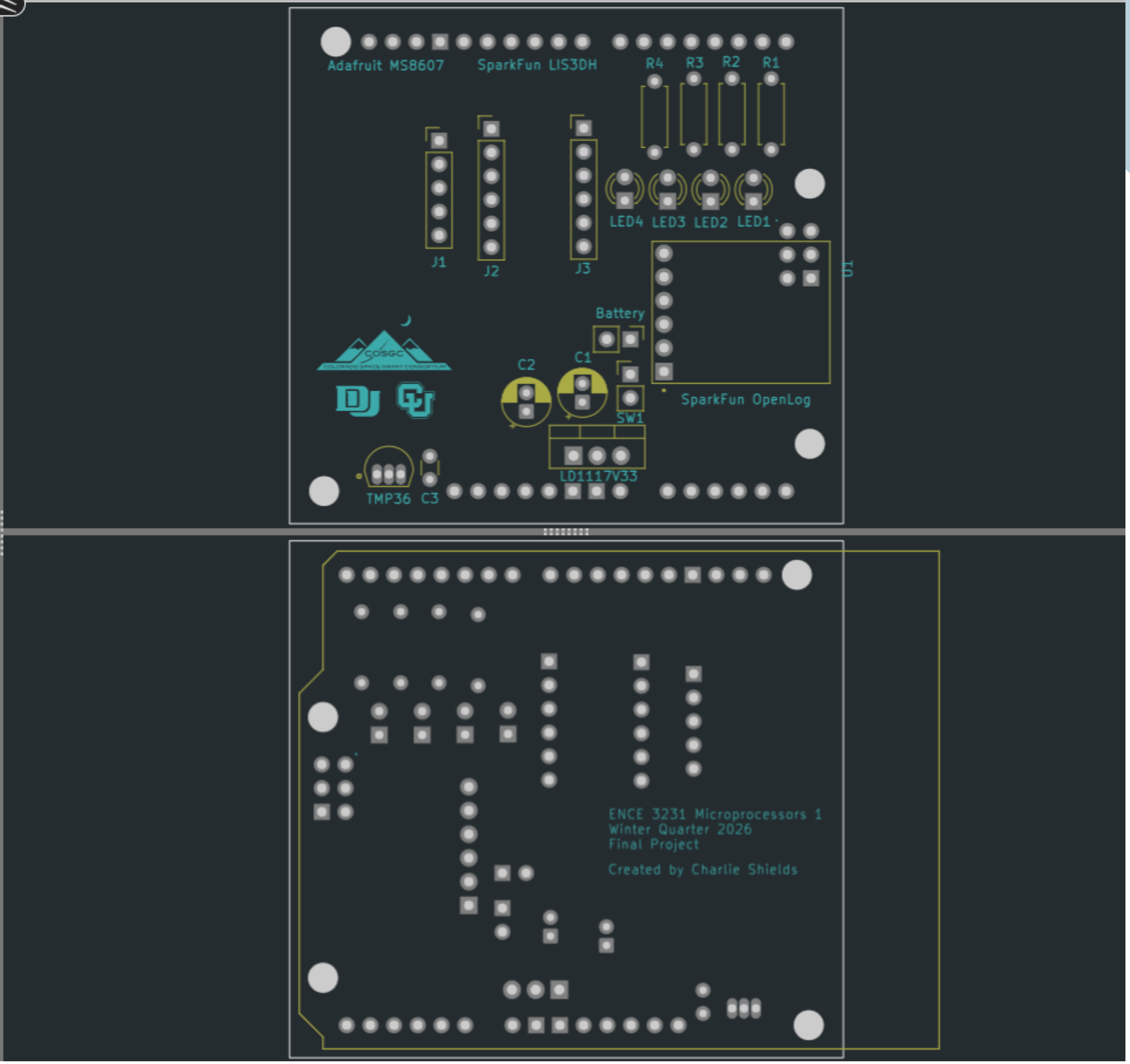
final_pcb

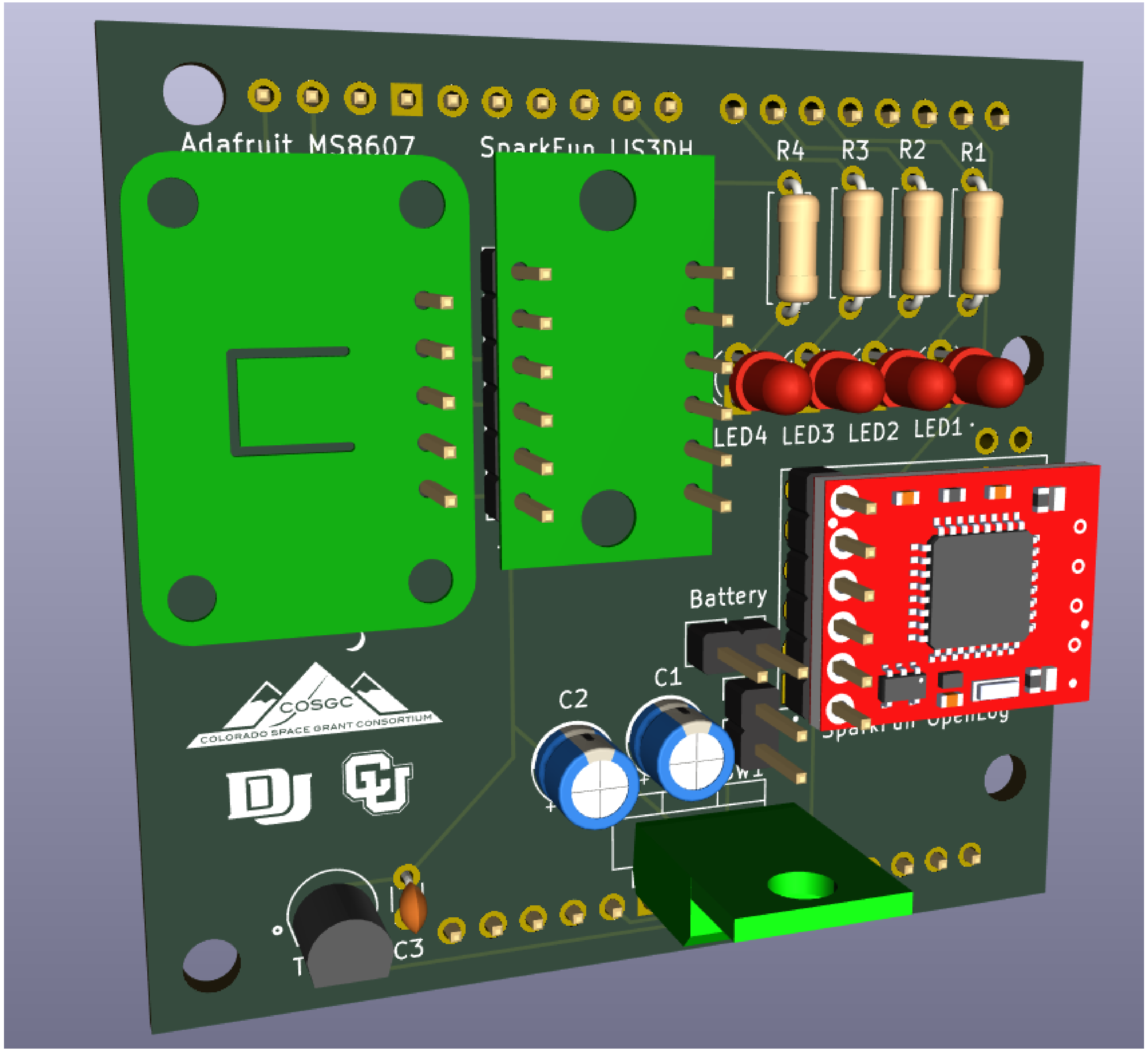
Rev:
2026-03-12 23:17:14

Ref lookup

Filter

	Sourced	Placed	References	Value	Footprint	Quantity
1	☐	☐	C1, C2	10µF	CP_Radial_D5.0mm_P2.00mm	2
2	☐	☐	C3	0.1uF	C_Disc_D3.0mm_W1.6mm_P2.50mm	1
3	☐	☐	R1, R2, R3, R4	470	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	4
4	☐	☐	D1	LED1	LED_D3.0mm	1
5	☐	☐	D2	LED2	LED_D3.0mm	1
6	☐	☐	D3	LED3	LED_D3.0mm	1
7	☐	☐	D4	LED4	LED_D3.0mm	1
8	☐	☐	U1	SparkFun OpenLog	MODULE_DEV-13712	1
9	☐	☐	U4	LD1117V33	TO-220-3_Vertical	1
10	☐	☐	U6	TMP36	XDCR_TMP36GT9	1
11	☐	☐	SW1	SW_SPST	PinHeader_1x02_P2.54mm_Vertical	1
12	☐	☐	A1	Arduino_Uno_R3_Shield	Arduino_Uno_R3_Shield	1
13	☐	☐	BT1	Battery	PinHeader_1x02_P2.54mm_Vertical	1
14	☐	☐	J2, J3	Conn_01x06_Pin	PinHeader_1x06_P2.54mm_Vertical	2
15	☐	☐	J1	Adafruit MS8607	PinHeader_1x05_P2.54mm_Vertical	1




Adafruit MS8607
R4
R3
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C1
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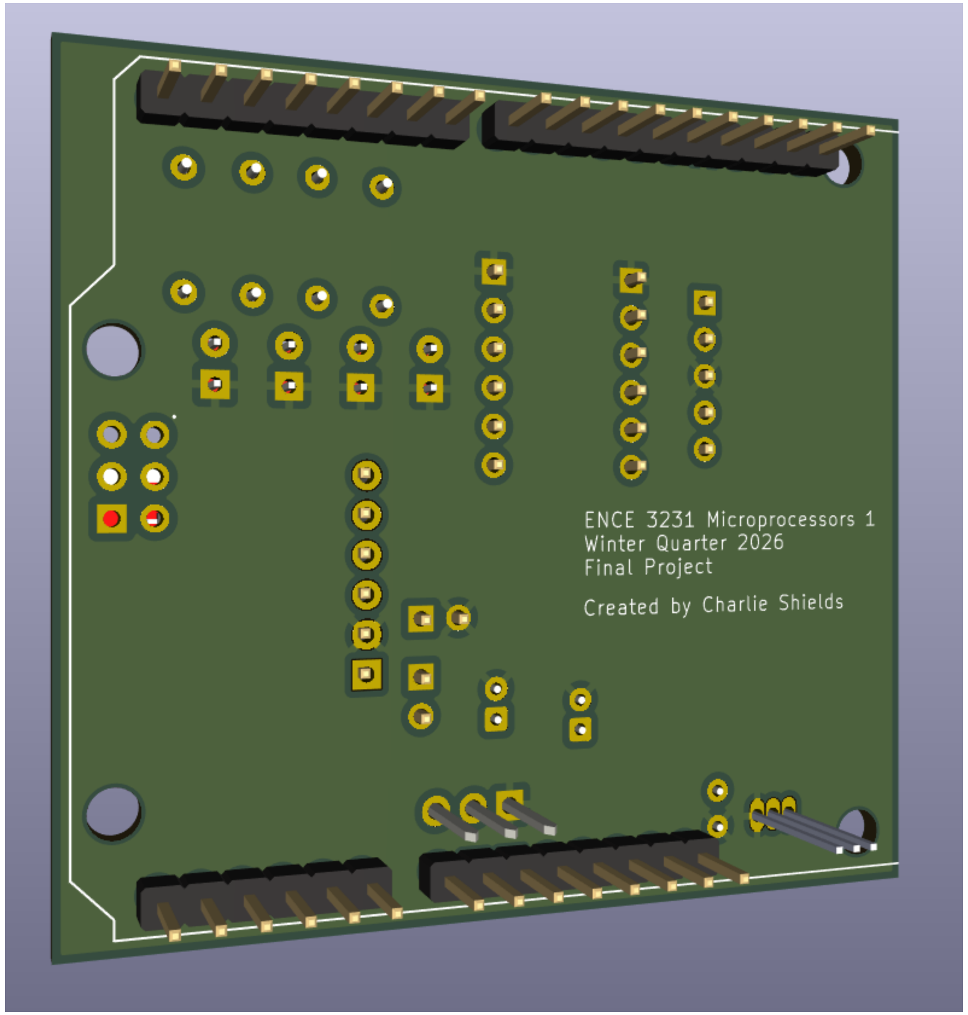
C3
COSGC
COLORADO SPACE GRANT CONSORTIUM

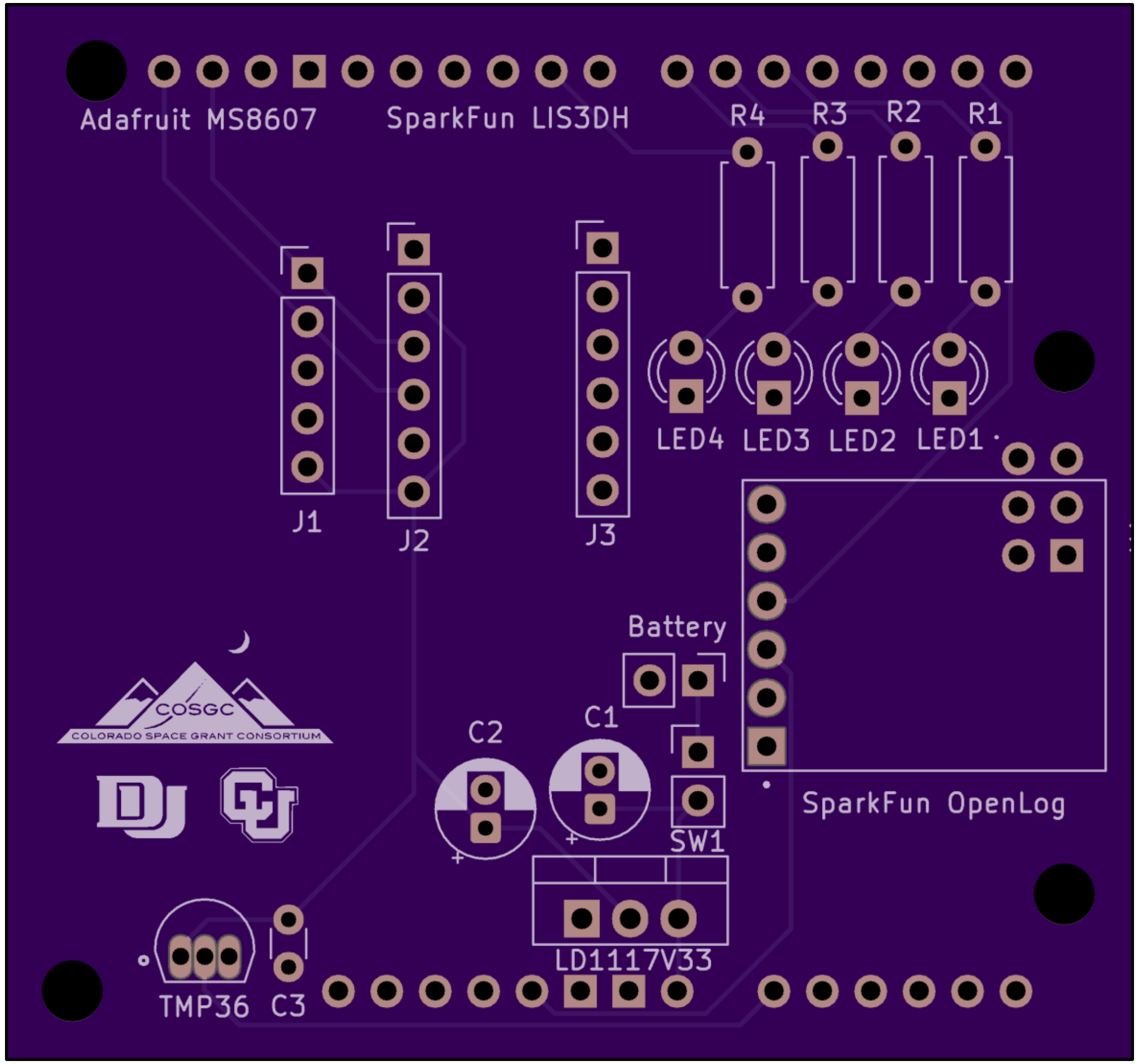

ENCE 3231 Microprocessors 1
Winter Quarter 2026
Final Project
Created by Charlie Shields

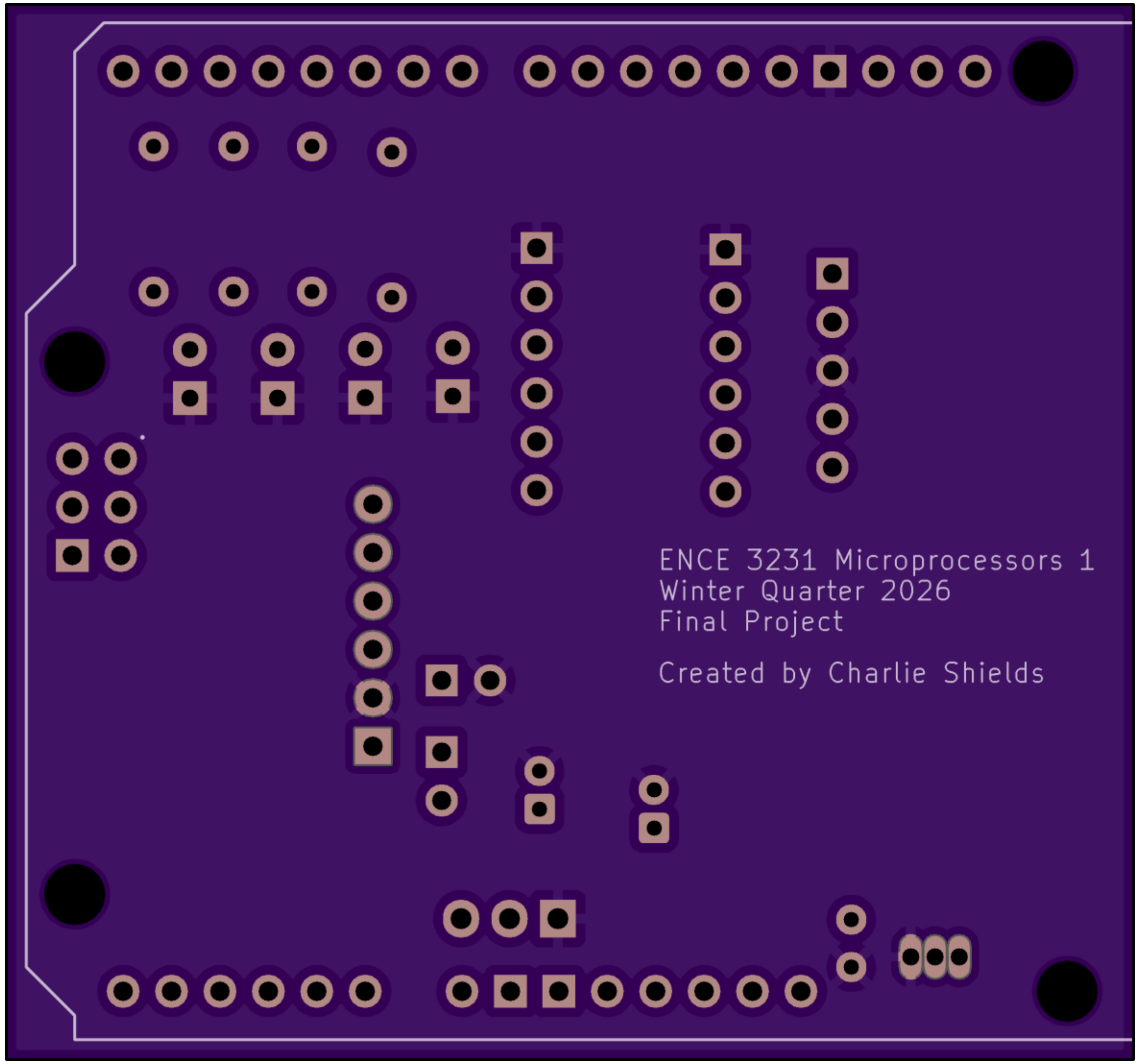
OSH-PARK

About your board

We detected a 2 layer board of 2.33 x 2.18 inches (59.1 x 55.4mm)

3 boards will cost $25.35





Phase 1 vs. Phase 2

Cost	Phase 1	Phase 2
Sensors	HIH-4030 ($53k!), ASDX ($35+), ADXL335 ($22.27)	MS8607 ($14.95), LIS3DH ($7.75)
Data logger	$17.50	$17.50
Total	**~$70+ (with unobtainable parts)**	**~$40.20 (all available)**

Weight	Phase 1	Phase 2
Electronics + PCB	~108-113 g	~82 g
Battery	~45 g	~45 g
Enclosure	~30-50 g	~30-50 g
Total	**~138-163 g**	**~112-127 g**

Power	Phase 1	Phase 2 (lin reg)	Phase 2 (buck)
Useful load	~308 mW	~275 mW	~275 mW
Regulator loss	~184 mW (hidden)	~570 mW	~50 mW
Total from battery	**~492 mW**	**~845 mW**	**~325 mW**
Est. battery life	~3 hrs	~1.5 hrs	~4-5 hrs

Improvements

- I2C Bus (Heavy)
- Linear Voltage Regulator (5.7v drop)